1. What is the worst-case time complexity of linear search algorithm?
(a) O(n) (b) O(i)
(c) O(logn) (d) $O(n^2)$
(CUET-PG 2021) Linear Search

2. Which of the following is not an application of binary search?
(a) To find the lower/upper bound in an ordered sequence
(b) union of intervals
(c) Debugging
(d) To search in unordered list
(CUET-PG 2021) Binary Search (Applications)

3. The complexity of binary search algorithm is
(a) O(n) (b) O(log n)
(c) $O(n^2)$ (d) O(n log n)
(CUET-PG 2021) Time Complexity

4. Given an array A = {55, 66, 77, 88, 99} and a key = 88. How many iterations are done until the element is found?
(a) 1 (b) 3
(c) 4 (d) 2
(CUET-PG 2021) Dry Run / Iteration Count

5. Which of the following operations is not O(1) for an array of sorted data? You may assume that array elements are distinct
(a) Find the ith largest element
(b) Delete an element
(c) Find the ith smallest element
(d) All of the above
(CUET-PG 2021) Time Complexity

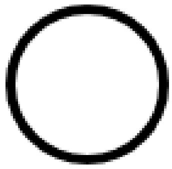
6. Quicksort is a
(a) greedy algorithm
(b) divide and conquer algorithm
(c) dynamic programming algorithm
(d) backtracking algorithm
(CUET-PG 2021) Quick Sort

7. Informal high-level description of an alrogithm in English is called
(a) function (b) class
(c) Pseudo code (d) None of the above
Computer – Pseudocode – 2021

8. What is the scope of the variable declared in the user-defined function?
(a) in whole program (b) only inside the {} block
(c) Both 1 and 2 (d) None of these
Computer – Scope of Variables – 2021

9. The time factor when determining the efficiency of an algorithm is measured by
(a) counting microseconds
(b) counting the number of key operations
(c) counting the number of statements
(d) counting the kilobytes of algorithm
Computer – Algorithm Analysis – 2021

10. The geometrical figure shown below in flowchart represents



(a) input
(b) connector
(c) decision
(d) looping
Computer – Flowchart Symbols – 2021

11. Which of the following is a structured programming technique that graphically represents the detailed steps required to solve a program?
(a) Algorithm (b) Pseudocode
(c) Flowchart (d) Top-down design
Computer – Flowchart – 2021

12. To evaluate an expression without any embedded function calls
(a) one stack is enough
(b) two stacks are needed
(c) as many stacks as the height of the expression tree are needed
(d) a Turing machine is needed in the general case
Computer – Stack – 2021

13. Which of the following data structures in a linear type?
(a) Graph (b) Tree
(c) Binary (d) Queue
Computer – Data Structures (Linear) – 2021

14. A graph is a collection of
(a) rows and columns (b) vertices and edges
(c) equations (d) matches
Computer – Graph Theory – 2021

15. The given numbers are inserted into an empty binary search tree in the given order: 10,1,3,5,15,12 and 16. What is the height of the binary search tree.
(a) 3 (b) 4
(c) 5 (d) 6
Computer – Binary Search Tree – 2021

16. Which of the following ways can be used to represent a graph?
(a)only Adjacency list (b) only adjacency Matrix
(c) only incidence Matrix (d) All of the above
Computer – Graph Representation – 2021

17. In a B* tree, If the search-key value is 8 bytes long, the block size is 512 bytes and the block pointer size is 2 bytes, then maximum order of the B* tree is
(a) 50 (b) 51
(c) 52 (d) 54

Computer – B-Tree / B*-Tree – 2021

18. The number of different binary trees with 6 nodes is
(a) 6 (b) 42
(c) 132 (d) 256

Computer – Binary Trees (Counting) – 2021

19. Which of the following standard algorithms is not a greedy algorithm?
(a) Prim's algorithm (b) Kruskal algorithm
(c) Dijkstra's shortest path algorithm
(d) Bellman-Ford shortest path algorithm

Computer – Algorithms (Greedy) – 2021

20. Prim's algorithm is a/an
(a) divide and conquer algorithm
(b) greedy algorithm
(c) dynamic programming
(d) approximation algorithm

Computer – Graph Algorithms (Prim's) – 2021

21. The following integers are needed to be sorted in ascending order using bubble sort.

5, 8, 22, 18, 1

Following are the results of various passes during the sorting process.
(A) 5,1,8,18,22 (B) 1,5,8,18,22
(C) 5,8,18,1,22 (D) 5,8,1,18,22
(a) C, D, B, A (b) C, D, A, B
(c) D, C, A, B (d) D, C, B, A

(CUET-PG 2024) Algorithms – Bubble Sort (Pass Analysis)

22. Arrange the following in the increasing order of their asymptotic complexities.
(A) Insertion sort (best case)
(B) Bubble sort (worst case)
(C) Binary search (worst case)
(D) Merge sort (worst case)
(a) (A), (C), (B), (D) (b) (D), (A), (C), (B)
(c) (A), (B), (C), (D) (d) (C), (A), (D), (B)

(CUET-PG 2024) Algorithm Analysis

23. Each node is having a successor node in ______________
(a)Singly linked list
(b) Singly circular linked list
(c)Doubly linked list
(d)Not possible in any linked list

Data Structures – Linked List (Circular Singly Linked List)

24. If we want to find last node of a singly linked list then the correct coding is
(a) If(temp→ link!=NULL) temp= temp→link
(b) If (temp→data = = Num) temp = temp → link
(c) While (temp → link ! = NULL) temp = temp → link
(d) While (temp →link ! = data) temp = temp → link

Data Structures – Linked List Traversal (Finding Last Node)

25. What are the ways to implement a priority Queue?
(A) Arrays (B) Fibonacci tree
(C) Heap Data Structure (D) Linked list
Choose the correct answer from the options given below
(a) A, B and D only (b) B, C and D only
(c) A, B, C and D (d) A, C and D only

Data Structures – Priority Queue Implementation

26. Which of the following is not an application of Stack?
(a) Tower of Hanoi (b) Recursion
(c) Voter polling station (d) Parantheses Matching

Stack – Applications

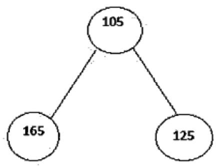
27. Consider implementation of a databse. Among the following options, choose the most appropriate data structure for this
(a) B+ tree (b) Linked list
(c) Queue (d) Stack

Data Structures – Indexing Structure (B+ Tree)

28. Consider a completely skewed (left/right) binary search tree with n elements. What is the worst case time complexity of searching an element in this tree?
(a) O(n) (b) O(1) (c) O(log n) (d)O (nlogn)

Data Structures – Binary Search Tree (Worst-Case Search)

29. Consider the following tree. This is a/an __________



(a)AVL Search Tree (b) Binary Tree
(c) Binary Search Tree (d) Fibonacci Tree

Trees – Binary Tree Identification

30. Consider the task of finding the shortest path in an unweighted graph by using BFS and DFS.

(CUET-PG 2025) Graph Theory (BFS vs DFS – Shortest Path)

Which of the following statements are true?
(A) BFS always finds the shortest path.
(B) DFS always finds the shortest path.
(C) DFS does not guarantee finding the shortest path.

(D) BFS does not guarantee finding the shortest path.
Choose the **correct** answer from the options given below:
(a) (B) and (D) only (b) (A) and (C) only
(c) (A) and (B) only (d) (C) and (D) only

31. Which of the following is not an application of DFS?
(CUET-PG 2025) DFS Application
(a) Topological
(b) Determining Strongly Connected Components is a graph
(c) Finding minimum distance to a node in an unweighted graph optimally
(d) Solving Maze Problem

32.

List-1 (Algorithm)		List-II (Recurrence relation)	
(A)	Binary Search	**(I)**	T(n) = T(n/2) + c (where c is a constant)
(B)	Merge Sort	**(II)**	T(n) = 2T(n/2) + ⨀(n)
(C)	Quick sort (worst case partitioning	**(III)**	T(n) = T(n – 1) + ⨀ (n)
(D)	Linear Search	**(IV)**	T(n) = T(n – 1) + c (where c is a constant)

(CUET-PG 2025) Recurrence relations
Choose the correct answer from the options given below:
(a) (A)(1), (B) (II), (C) - (III), (D) - (IV)
(b) (A)(1), (B) (III), (C) (II), (D) - (IV)
(c) (A) (1), (B) (II), (C) (IV), (D) - (III)
(d) (A) (III), (B) (IV), (C) (I), (D) - (II)

33. The Quicksort and randomized Quicksort procedures differ in:
(CUET-PG 2025) Algorithms
(a) Selection of Pivot element
(b) Worst case time complexity
(c) Best case time Complexity
(d) Final Output

34. Given the index i of a node in a heap, we can not computer:
(CUET-PG 2025) Heap Properties
(a) Parent (i) (b) Heap size
(c) Left (i) (d) Right (i)

35. In a binary search tree, the worst case time complexity of inserting and deleting a key is:
(CUET-PG 2025) Time Complexity
(a) O(log n) for insertion and O(log n) for detetion
(b) O(n) for insertion and O(log n) for detetion
(c) O(n) for insertion and O(n) for detetion
(d) O(log n) for insertion and O(n) for detetion

36. Arrange the following time complexities in increasing order. (CUET-PG 2025) Time Complexity
(a) Bubble sort (worst case)
(b) Deleting head node in singly linked list
(c) Binary search
(d) Worst case of merge sort
Choose the correct answer from the options given below:
(a) (A), (B), (C), (D) (b) (B), (C), (D), (A)
(c) (B), (A), (D), (C) (d) (C), (B), (D), (A)

37. In case of Binary search tree which of the following procedure's running time is distinct among all
(CUET-PG 2025) Bst Operation Time
(a) TREE-SUCCESSOR
(finds successor of the given node)
(b) TREE-MAXIMUM
(finds the node with maximum value)
(c) INORDER-WALK
(prints all elements of a tree in inorder manner)
(d) TREE-MINIMUM
(finds the node with minimum value)

ANSWER KEY

1.	2.	3.	4.	5.
a	d	b	b	b
6.	7.	8.	9.	10.
b	c	b	b	b
11.	12.	13.	14.	15.
c	a	d	b	a
16.	17.	18.	19.	20.
d	b	c	d	d
21.	22.	23.	24.	25.
c	d	b	c	d
26.	27.	28.	29.	30.
c	a	a	b	b
31.	32.	33.	34.	35.
c	a	a	b	c
36.	37.			
b	c			